

EDUCATION

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science and Engineering (AI) GPA: 7.52 / 10

– Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning

Chennai, India

Aug. 2023 – May 2027

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C++

Frontend: React.js, Next.js, React Native, Vite, Tailwind CSS, Responsive UI Design

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT Authentication, OAuth 2.0

Databases: PostgreSQL, MongoDB, MySQL, Redis

DevOps & Cloud: Docker, Git, GitHub Actions, CI/CD, AWS (S3, EC2), Linux

Data & Streaming: Apache Kafka, Apache Spark, Hadoop, HDFS, Real-Time Analytics

AI & LLM: Ollama, LangChain, OpenAI API, OCR, Prompt Engineering

PROJECTS

Focus Flow AI: Productivity Intelligence Platform

Next.js, TypeScript, FastAPI, Ollama (LLaMA 3), Firebase, Docker, OAuth 2.0 

– Architected a full-stack AI productivity platform integrating Google Calendar and Gmail APIs via OAuth 2.0; built a FastAPI backend delivering LLM-driven cognitive load inference via Ollama (LLaMA 3) at sub-300ms API response latency.

– Engineered a Dockerized multi-service stack (Next.js frontend, FastAPI server, LLM inference); implemented Firebase real-time sync for session and concurrent multi-user workflow management across the platform.

Dec 2025 – Jan 2026

Smart-City Real-Time Traffic Intelligence System

Apache Kafka, Spark Streaming, Hadoop, HDFS, MongoDB, Python, scikit-learn 

– Architected a distributed streaming pipeline ingesting 10,000+ IoT events/sec via Apache Kafka with Spark Streaming jobs for real-time congestion classification, reducing event-to-alert latency by 40% over batch baselines across a 5-tier severity model.

– Designed an HDFS-backed data lake with MongoDB operational storage; trained ML classifiers achieving 89% congestion prediction accuracy; exposed insights via a REST API layer serving live monitoring dashboards.

2025

E-Commerce Legal Metrology Compliance Platform

React (Vite), TypeScript, Node.js, Express.js, MySQL, Docker, Tesseract OCR 

– Developed a production-grade full-stack compliance system with a React/TypeScript SPA and Node.js/Express REST API; integrated Tesseract OCR achieving 94% field-accuracy, validating 500+ product listings against 15+ legal metrology regulations.

– Implemented RBAC, PDF report export, and a normalized MySQL schema; containerized the full stack with Docker, enabling reproducible deployment and streamlined development-to-staging CI workflows.

Aug – Sep 2024

PUBLICATIONS (C = Conference)

[C.1] Subhash R. (2024). **QR Code Encryption Using LU Decomposition and PCA**. *IConSCEPT 2024*, IEEE. **Best Research Paper Award**. DOI: 10.1109/IConSCEPT61884.2024.10627798

[C.2] Subhash R. (2025). **GAM Oversampling and GMM-Based Resampling for Imbalanced Credit Data Classification**. *IEEE InC 2025*. DOI: 10.1109/InC465408.2025.11256257

ACHIEVEMENTS

Best Research Paper Award | *IConSCEPT 2024, IEEE International Conference, NIT-PY, India*

– Recognized for original cryptographic research among 200+ submitted papers across IEEE signal processing tracks.

2024

2nd Place — Coders of the Black Pearl | *Neuroinx Club (Karur Vysya Bank), Amrita Vishwa Vidyapeetham*

– Secured 2nd place among 80+ participants demonstrating algorithmic problem-solving under competitive time constraints.

2024

Top 4 Finalist — Shaastra Robosoccer | *Shaastra 2025, IIT Madras*

– Placed among the top 4 of 100+ national university teams; built autonomous navigation and real-time object-tracking systems.

Jan 2025

CERTIFICATIONS & MEMBERSHIPS

ML, Data Science & AI Engineering with Python — Udemy

Machine Learning Certification — Corizo (Jul 2024)

IEEE Member, ID: 100981642 (Jan 2025 – Present)

TensorFlow Deep Learning Bootcamp — Udemy

Other Certifications — View Credentials